GitHub — Zero to Master

22 topics, grouped as Beginner → Intermediate → Advanced → Master. Work through the topics in order; type every code example yourself and finish the practice tasks.

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Beginner Level

01. Introduction to GitHub

GitHub hosts Git repositories and provides collaboration tools: issues, pull requests, Actions, Pages and Projects.

02. Creating an Account and Repository

Create a repo, add a README, license and `.gitignore`; choose public or private.

```
git remote add origin https://github.com/user/repo.git
git push -u origin main
```

03. Authentication: SSH and Tokens

Use SSH keys or fine-grained personal access tokens; enable 2FA.

```
ssh-keygen -t ed25519 -C "you@example.com"
```

04. Cloning, Forking and Pushing

Clone to work locally; fork to contribute to repos you do not own.

```
git clone https://github.com/user/repo.git
```

05. Issues and Labels

Track bugs and tasks; templates, labels, assignees, milestones.

06. README and Markdown

Write a clear README with badges, setup, usage and screenshots using Markdown.

````
# Project
## Install
```bash
npm i
```
````

Intermediate Level

07. Pull Requests and Code Review

Branch, push, open a PR, request reviewers, resolve comments, squash/merge.

08. Branch Protection and Rulesets

Require reviews, status checks and linear history on `main`.

09. Collaboration: Teams, Permissions, CODEOWNERS

Organisations, teams, roles, and CODEOWNERS for automatic reviewers.

```
# .github/CODEOWNERS
/src/ @team-frontend
```

10. GitHub Projects

Plan with boards, tables and roadmaps linked to issues.

11. GitHub Pages

Host static sites directly from a branch or Actions.

12. Releases and Tags

Publish versioned releases with notes and binaries.

```
gh release create v1.0.0 --generate-notes
```

Advanced Level

13. GitHub Actions Basics

Workflows in `.github/workflows/*.yml` with events, jobs and steps.

```yaml
name: CI
on: [push, pull_request]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: npm ci && npm test
```

14. Actions: Matrix, Cache, Artifacts, Secrets

Test multiple versions, cache dependencies, share artifacts and use encrypted secrets.

```yaml
strategy:
  matrix:
    node: [20, 22]
```

15. Deploying with Actions

Deploy to Pages, cloud providers, containers using OIDC instead of long-lived secrets.

16. Reusable Workflows and Custom Actions

Share logic across repos with `workflow_call` and composite/JS actions.

```yaml
uses: org/repo/.github/workflows/ci.yml@main
```

17. Security: Dependabot, Code Scanning, Secret Scanning

Automate dependency updates and vulnerability detection; push protection for secrets.

```yaml
# .github/dependabot.yml
updates:
  - package-ecosystem: npm
```

18. GitHub CLI and API

Automate with `gh` and the REST/GraphQL APIs.

```
gh pr create --fill
gh issue list
```

Master Level

19. Open Source Contribution Workflow

Find good-first-issues, fork, branch, follow CONTRIBUTING, write tests, open a clear PR.

20. Repository Governance at Scale

Rulesets, required workflows, audit logs, SSO/SAML, Enterprise policies.

21. Monorepo and Release Automation

Semantic versioning, changesets/release-please, automatic changelogs.

22. Real Project: Portfolio + CI/CD

A portfolio site on GitHub Pages with tests, lint, preview deployments and Dependabot.

Interview Question Bank

Q1. Fork vs clone?
A fork is your server-side copy of someone else's repo; a clone is a local copy of any repo.

Q2. What is a pull request?
A request to merge a branch, with review, discussion and checks.

Q3. What is GitHub Actions?
GitHub's built-in CI/CD and automation platform driven by workflows.

Q4. What are GitHub Pages?
Free static hosting for repositories.

Q5. How do you protect the main branch?
Branch protection/rulesets: required reviews, passing checks, no force pushes.

Q6. What is Dependabot?
A bot that opens PRs to update vulnerable or outdated dependencies.

Q7. Secrets in Actions?
Encrypted repository/environment secrets exposed as `${{ secrets.NAME }}`.

Q8. Squash vs rebase vs merge commit?
Squash combines commits into one; rebase replays them linearly; merge keeps full history with a merge commit.

Q9. What is CODEOWNERS?
A file assigning default reviewers to paths.

Q10. What is a GitHub Release?
A tagged, shareable version with notes and attached assets.

Capstone Project Ideas

1. Portfolio site on Pages
2. CI workflow for a Node/Python project
3. Automated release workflow
4. Open-source contribution
5. Dependabot + code scanning setup